⌜Def.

A group $D_{2n} \overset{\text{def}}{=} \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$ is the dihedral group of order $2n$.

⌜Prop. Let D_{2n} be the dihedral of order $2n$.

1). For any $x \in D_{2n} \setminus \langle r \rangle$, $rx = xr^{-1}$.

2). For any $x \in D_{2n} \setminus \langle r \rangle$, $|x| = 2$.

3). $D_{2n} = \langle s, sr \rangle$.

4). If $n = 2k$ is even with $n \geq 4$, then r^k has order 2 and $\langle r^k \rangle = \{1, r^k\} = Z(D_{2n})$.

5). If n is odd with $n \geq 3$, then $Z(G) = \{1\}$.

Proof. Let $x \in D_{2n}$ be given. Then $x = r^a s$ for some $a = 0, 1, \dots, n-1$, by $x \notin \langle r \rangle$.
So, $rx = r^{a+1} s = r^a \cdot r \cdot s = r^a s r^{-1} = x \cdot r^{-1}$.
And, $x^2 = r^a s r^a s = r^{a-1} s r^{-1} r^a s = \cdots = s r^{-a} \cdot r^a \cdot s = s^2 = 1$, and $x \neq 1$, so $|x| = 2$. ⌟ The 3) is given by $\langle s, sr \rangle \subseteq D_{2n}$ and $ssr = r \in \langle s, sr \rangle$, so $\langle s, sr \rangle \supseteq \langle s, r \rangle = D_{2n}$. ⌟

To determine the center of D_{2n}, observe that; given $a \in \{0, 1, \dots, n-1\}$,

$$r^a \in Z(D_{2n}) \iff r^a s = s r^a = s r^{-a} \iff r^{2a} = 1,$$ by the cancelation law.

Since $r^{2a} = 1$ if and only if $n \mid 2a$, so $r^a \in Z(D_{2n})$ if and only if

$$a = \begin{cases} 0 & \text{if } n \text{ is odd} \\ 0, \frac{n}{2} & \text{if } n \text{ is even.} \end{cases}$$

And, given $x \in D_{2n} \setminus \langle r \rangle$, $x \in Z(D_{2n})$ implies that $xr = rx = xr^{-1} \Rightarrow r^2 = 1$.
But, it contradicts to $|r| = n \geq 3$. Consequently, for $n \geq 3$,

$$Z(G) = \begin{cases} \{1\} & \text{if } n \text{ is odd} \\ \{1, r^k\} & \text{if } n = 2k \text{ is even} \end{cases}$$

☑

「Prop. If n is even, then $D_{2n} \cong \langle (1\ 2 \cdots n), (2\ n)(3\ n-1)\cdots(\frac{n}{2}\ \frac{n}{2}+2) \rangle \leq S_n$

If n is odd, then $D_{2n} \cong \langle (1\ 2 \cdots n), (2\ n)(3\ n-1)\cdots(\frac{n+1}{2}\ \frac{n+1}{2}+1) \rangle \leq S_n$.

Proof In case of n is even number, observe that

$$(1\ 2 \cdots n)(2\ n)(3\ n-1)\cdots(\tfrac{n}{2}\ \tfrac{n}{2}+2)(1\ 2 \cdots n)$$
$$=(1)(2\ n)(3\ n-1)\cdots(\tfrac{n}{2}\ \tfrac{n}{2}+2)$$

Moreover, $(1\ 2 \cdots n)^n = id$ is clear, and since $(2\ n)\cdots(\frac{n}{2}\ \frac{n}{2}+2)$ is product of disjoints two-cycles, so it has order 2.

So, corresponding to $r = (1\ 2 \cdots n)$ and $s = (2\ n)\cdots(\frac{n}{2}\ \frac{n}{2}+2)$, it satisfies the representation of the definition. Similarly, n is odd, can shown. ☒

D_6.

By the theorem above, $D_6 \cong S_3$. Now, explicitly,

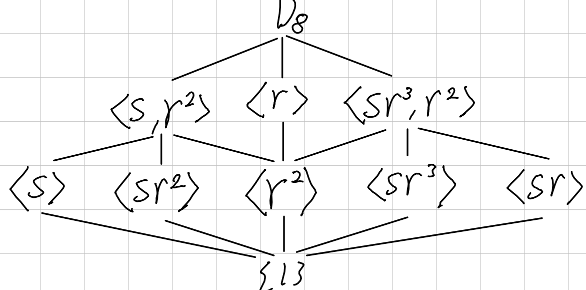
$$\begin{tikzcd}[column sep=small]
 & & D_6 \arrow[dll, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\langle r\rangle \arrow[drr, dash, "3"'] & & \langle s\rangle \arrow[d, dash, "2"'] & \langle sr\rangle \arrow[dl, dash, "2"] & \langle sr^2\rangle \arrow[dll, dash, "2"] \\
 & & \{id\} = Z(D_6). & &
\end{tikzcd}$$

Normalizer:

$$N_{D_6}(\langle r\rangle) = D_6$$

D_8

$D_8 = \langle r, s \mid r^4 = s^2 = 1,\ rs = sr^{-1} \rangle$



Normalizer: Since index of 2 subgroups must be normal, the normalizer of $\langle s, r^2 \rangle$, $\langle r \rangle$, $\langle sr^3, r^2 \rangle$ is just D_8.
$N_{D_8}(\langle s \rangle) = \langle s, r^2 \rangle$, because: Since $|\langle s, r^2 \rangle : \langle s \rangle| = 2$, $\langle s, r^2 \rangle \leq N_{D_8}(\langle s \rangle)$, but

$$r \langle s \rangle r^{-1} = \{1, sr^2\} \neq \langle s \rangle, \text{ so } N_{D_8}(\langle s \rangle) \neq D_{2n}.$$

Similarly $N_{D_8}(\langle sr^2 \rangle) = \langle s, r^2 \rangle$. But, $N_{D_8}(\langle r^2 \rangle) = D_8$, because:
first way: Since $|\langle s, r^2 \rangle : \langle r^2 \rangle| = |\langle r \rangle : \langle r^2 \rangle| = 2$, therefore

$$\langle s, r^2 \rangle, \langle r \rangle \subseteq N_{D_8}(\langle r \rangle), \text{ so } N_{D_8}(\langle r \rangle) = D_8.$$

Alternatively, since $sr^2s^{-1} = sr^2s = r^{-2}s \cdot s = r^2$, so $s, r \in N_{D_8}(\langle r \rangle)$, so $\langle s, r \rangle \subseteq N_{D_8}(\langle r \rangle)$. ($r \in D_8$ clearly normalizes $\langle r^2 \rangle$).

Centralizer: First, $C_{D_8}(D_8) = \{1, r^2\} = \langle r^2 \rangle$ because: $sr^2 = r^{-2}s$ and clearly r^2 commutes all elements in $\langle r \rangle$, so r^2 commutes all elements of D_8. but, $sr^3 = rs \neq r^3 s$, so $r, s \notin Z(G)$ and $sr^2 \notin Z(G)$ because $sr^2 \in Z(G) \Rightarrow sr^2 r^{-2} = s \in Z(G)$ since the center is a subgroup.
Moreover, $sr \cdot s = r^3 ss = r^3 \neq ssr = r$, so $sr \notin Z(G)$. Thus $Z(G) = \langle r \rangle$
Next, $C_{D_8}(\langle r \rangle) = \langle r \rangle$, because: since $rs = sr^3 \neq sr$, so $s \notin C_{D_8}(\langle r \rangle)$ and clearly all elements in $\langle r \rangle$ commute with the all elements in $\langle r \rangle$, so $\langle r \rangle \subseteq C_{D_8}(\langle r \rangle) \subsetneq D_8$, so the Lagrange's theorem gives the result.

Commutator: Since the derived set is actually normal subgroup and it induces the largest abelian quotient, so $D_8' = \langle r^2 \rangle$, because $D_8 / \langle r^2 \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, because it contains two elements order 2, but it is not cyclic.

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